

Closed-Form Compression of the Sangbun Vector

Source text · Complete derivation for all cases · Reformulation in rotated coordinates ·
Verification

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Revised edition — see the Erratum on the following page

Abstract

In Sangbun theory the Sangbun vector AA_{12} has until now required walking through twelve construction steps, A_1 through A_{12} , in strict order. This document first reproduces the author's source text verbatim, then compresses the entire chain into closed-form expressions that require no line-intersection solving whatsoever. Five results stand out.

- The nine points A_{3a} through A_{5b} collapse into a single scalar λ .
- A'_{10} need never be computed; a single sign replaces it.
- The branch value $\mathfrak{Q}$ is obtained without any square root as $\mathfrak{Q} = |b|/|t|$, and further as $\mathfrak{Q} = |\tan \theta|$ for a single angle θ .
- In the rotated frame (τ, β) that lays the tangent flat, steps A_7 through A_{12} coincide exactly with the $Z2$ ($y = c$) formulas. That is, everything after A_7 is rotation-invariant and does not depend on the slope a .
- The Pythagorean relation $|OA|^2 = |AA_1|^2 + |OA_1|^2$ turns out to be the skeleton of the whole system.

All cases are covered: $Z1$ ($y = ax + b$), $Z2$ ($x = k$ and $y = c$), $\mathfrak{Q} < 1$, $\mathfrak{Q} > 1$, $\mathfrak{Q} = 1$, $A = A_1$, and both signs of a . Results are checked against four existing construction plots to five decimal places, and the conditions of plot ② are computed in full.

Erratum — changes in this revised edition

The previous edition gave the closed form of AA_6 in §3.5 and in Table 2 as $(\tau + \lambda\beta)T + (2\beta - \lambda\tau)E$. *The sign of the E -component was wrong.* In the rotated frame one has $A_2 = (\tau, 0)$ and $A_1 = (0, \beta)$, hence $A_2^* = 2A_2 - A_1 = (2\tau, -\beta)$ and $A_2^* - A = (\tau, -2\beta)$. The correct expression is

$$AA_6 = (\tau + \lambda\beta)T + (\lambda\tau - 2\beta)E.$$

With the old sign, plots ② and ④ of Table 3 are not reproduced; with the corrected sign every entry of Table 3 agrees to five decimal places, and the closed form matches the full A_3 – A_5 construction to floating-point rounding (Table 4). The scalar λ itself (§3.3, §3.4) was and remains correct.

Other changes: §2.1 now states $A_2 = (\tau, 0)$ explicitly, which is what fixes the sign; Tables 4, 5 and the residuals of §7.4 have been recomputed with the corrected formula; Table 5 has been extended from seven to ten parameter combinations; and §7.6 adds a full step-by-step computation of plot ② together with its figure.

0. Source Text — Sangbun Vector Computation and Plotting (authoritative)

Chapter 0 reproduces the author’s source text. Every derivation in the later chapters takes this specification as its sole starting point; nothing outside it is assumed. Plotting conventions (labels, connecting segments, axis ratios) are irrelevant to the computation and are summarised.

General remarks

Sangbun theory is a geometric-vector methodology that interprets, step by step, the geometric structure defined at a single point and the vector relations generated around that point. Tangent, normal, projection, rotation and vector addition are geometric processes that all originate at the point and link to one another; the theory explores, through figure-based intuitive interpretation, how these together form one vector structure.

The Sangbun vector is a double-structure vector combining a straight-line geometric component generated by the tangent-projection structure with a rotational geometric component generated by the circle-rotation structure.

Computing a vector at a point on a line or curve. “The construction below assumes non-degenerate general position.” “Special cases in which an intersection is undefined are defined by separate rules.”

[Z1] Basic setup

slope: a origin: $O(0,0)$ tangent: $y = ax + b$ point: $A(x, ax + b)$

Basic projections and intersections

- A_1 : foot of the perpendicular from O to the tangent
- A_2 : foot of the perpendicular from O to the normal at A
- A_{3a} : intersection of the normal at A with the x -axis; $A'_{3a} = (A_{3a}, A_{1y})$
- A_{3b} : intersection of the normal at A with the y -axis; $A'_{3b} = (A_{1x}, A_{3b})$
- A_{3c} : intersection of the vertical line at A_{3a} with the tangent
- A_{3d} : intersection of the horizontal line at A_{3b} with the tangent

Feet of perpendiculars relative to the origin

- A_{4a} : intersection of line OA_{3c} with line AA'_{3a}
- A_{4b} : intersection of line OA_{3d} with line AA'_{3b}

Construction of A_6 (vector sum)

- A_{5a} : intersection of line (1) — the line through A_2 perpendicular to A_1A_2 — with the extension of A_1A_{4a}
- A_{5b} : intersection of line (1) with the extension of A_1A_{4b}
- Definition of A_6 : $A_1A_6 = A_1A_{5a} + A_1A_{5b}$

Circle (1) and principal points

- circle (1): centre A_1 , radius $|A_1A|$
- A_7 : the point where the ray from O in the direction of A_1 , continued past A_1 , meets circle (1)
- A_8 : the intersection of circle (1) with the line through A perpendicular to OA , other than A

- A_9 and A_{9a} : of the two points lying at distance $|A_7A_8|/2$ perpendicular to the midpoint of segment A_7A_8 , the one inside circle (1) is A_9 , the one outside is A_{9a}

Branch value $\mathcal{Q} = |OA_1|/|AA_1|$

Case $\mathcal{Q} < 1$. circle (2): centre A_8 , radius $|A_8A_9|$. A_{10} : the intersection of circle (2) with the line through A_9 perpendicular to AA_9 , other than A_9 . A'_{10} : intersection of the extension of AA_9 with the line through A_{10} having the reciprocal slope of AA_{10} . A_{11} : the point taken from A in the direction from A'_{10} toward A_{10} , with $|AA_{10}| = |AA_{11}|$.

Case $\mathcal{Q} > 1$. circle (2): centre A_9 , radius $|A_9A_7|$. A_{10} : the intersection of circle (2) with the line through A_7 perpendicular to AA_7 , other than A_7 . A'_{10} : intersection of the extension of AA_7 with the line through A_{10} having the reciprocal slope of AA_{10} . A_{11} : the point taken from A in the direction from A'_{10} toward A_{10} , with $|AA_{10}| = |AA_{11}|$.

Case $\mathcal{Q} = 1$. (1) $A_X > A_{1X}$, $A_Y < A_{1Y}$; (2) $A_X < A_{1X}$, $A_Y > A_{1Y}$; (3) $A_X > A_{1X}$, $A_Y > A_{1Y}$; (4) $A_X < A_{1X}$, $A_Y < A_{1Y}$. “ X is the point on line AA_7 selected by computation so as to satisfy the inequality conditions above.” Here $A_7 = A_8 = A_9 = A_{9a} = A_{10} = A'_{10}$, and A_{11} is the point obtained by rotating AA_7 through 90 degrees in the same sense in which AO rotates toward the direction AA_7 by at most 90 degrees, with $|AA_7| = |AA_{11}|$.

Final point A_{12} : $AA_{12} = AA_6 + AA_{11}$.

Special case: if $y = ax$ and $A(x, ax)$ then $A_{12} = A$.

[Z2] Definition and computation

The case in which the slope a is undefined. The reference line is $x = k$ or $y = c$. The point A takes one of four forms: $x = k$, $A = (k, k_1)$ / $y = c$, $A = (c_1, c)$ / $x = k$, $A = (k, k)$ / $y = c$, $A = (c, c)$. Origin $O = (0, 0)$.

- A_1 (always the same): foot of the perpendicular from O to the reference line. $x = k \rightarrow A_1 = (k, 0)$; $y = c \rightarrow A_1 = (0, c)$.
- A_2 (axis projection of A): for $x = k$, $A = (k, k_1)$ one has $A_2 = (0, k_1)$; for $y = c$, $A = (c_1, c)$ one has $A_2 = (c_1, 0)$.
- circle (1): centre A_1 , radius $|A_1A|$.
- A_6 : intersection of the line through A_2 perpendicular to A_1A_2 with the vertical/horizontal line of the reference line. $x = k \rightarrow$ vertical line, $y = c \rightarrow$ horizontal line.
- Definition of $\mathcal{Q}$ (same as $Z1$): $\mathcal{Q} < 1$, $\mathcal{Q} > 1$, $\mathcal{Q} = 1$; the subsequent steps apply the same branch rules as $Z1$.
- Computation of $A_7 \sim A_{12}$ (key rule): in $Z2$ the computation from A_7 onward is carried out in “exactly the same way” as in $Z1$. Only the reference line differs; the rotation, circle (2) and vector-sum rules are all exactly as in $Z1$.
- Special consequence in $Z2$: if $y = ax$ and $A = (x, ax)$ then $A_{12} = A$ (fully self-returning structure).

Plotting conventions (summary). Mark the origin $O(0,0)$; axis ratio 1:1; grid spacing 1. Coordinates are printed only for A , A_1 , A_{12} and A'_{10} , to five decimal places. The values of a , b , x , $\mathcal{Q}$ together with $|AA_{12}|$ and the angle of AA_{12} are printed at one side of the plot. A_{3a} , A'_{3a} , A_{3b} , A'_{3b} , A_{3c} and A_{3d} are not shown. When $\mathcal{Q} = 1$, circle (2) is undefined; only A'_{10} is shown and the connecting segment is AA'_{10} .

1. Notation and Basic Quantities

We start from the point $A = (p, ap + b)$ on the reference line $y = ax + b$. The origin is $O = (0, 0)$.

$$C = a^2 + 1, \quad H = \frac{b}{C}, \quad t = Cp + ab, \quad S = p^2 + (ap + b)^2.$$

Only three vector operations are used throughout.

$$\text{rot90}(x, y) = (-y, x), \quad u \cdot v = u_x v_x + u_y v_y, \quad \text{cross}(u, v) = u_x v_y - u_y v_x.$$

1.1 A_1 and A_2

Following the source text, A_1 is the foot of the perpendicular from O to the tangent and A_2 is the foot of the perpendicular from O to the normal at A . Since the normal has direction vector $(a, -1)$, both points follow from a single projection.

$$A_1 = H(-a, 1), \quad A_2 = A + H(a, -1) = (p + aH, a(p + aH))$$

Three identities follow at once. They form the skeleton of everything that comes later.

$$\begin{aligned} \text{(i)} \quad & A - A_1 = \frac{t}{C}(1, a) \Rightarrow |AA_1| = |t|/\sqrt{C} \\ \text{(ii)} \quad & |OA_1| = |b|/\sqrt{C} \\ \text{(iii)} \quad & |A_1 A_2| = |OA| \end{aligned}$$

Identity (i) says that $A_1 A$ is always parallel to the tangent direction $(1, a)$, and that its length is $|t|/\sqrt{C}$. Dividing (ii) by (i) gives the branch value immediately.

$$\mathfrak{Q} = \frac{|OA_1|}{|AA_1|} = \frac{|b|}{|t|}$$

The square root cancels. The definition in the source text is a ratio of two lengths, each carrying a factor $\sqrt{C}$; these cancel, leaving only a comparison between $|b|$ and $|t|$. Thus $\mathfrak{Q} < 1 \Leftrightarrow |b| < |t|$ and $\mathfrak{Q} = 1 \Leftrightarrow |b| = |t|$.

2. The Rotated Frame (τ, β)

The coordinate frame introduced in this chapter is the key to every compression that follows. It is the rotation that lays the tangent flat, and within it most of the Sangbun construction takes a far simpler form.

2.1 Definition

Put $\varphi = \arctan a$ and take unit vectors along the tangent and its perpendicular.

$$T = \frac{(1, a)}{\sqrt{C}} \quad (\text{tangent unit vector}), \quad E = \text{rot90}(T) = \frac{(-a, 1)}{\sqrt{C}}$$

Then define two signed lengths.

$$\tau = \frac{t}{\sqrt{C}} \quad (\text{signed } |AA_1|), \quad \beta = \frac{b}{\sqrt{C}} \quad (\text{signed } |OA_1|)$$

In the (T, E) frame four points become extremely simple.

$$O = (0, 0), \quad A_1 = (0, \beta), \quad A = (\tau, \beta), \quad \boxed{A_2 = (\tau, 0)}$$

That is, the tangent becomes the horizontal line $y = \beta$, A_1 is its foot on the vertical axis, and A sits τ to the side of it. This is exactly the configuration “ $y = c$, $A = (c_1, c)$ ” of $Z2$ in the source text. That fact becomes decisive in Chapter 4.

The statement $A_2 = (\tau, 0)$ is new in this edition and is worth deriving, since it is what fixes the sign in §3.5. From $(a, -1) = -\sqrt{C}E$ and $H\sqrt{C} = \beta$ we get $A_2 = A + H(a, -1) = A - \beta E$, i.e. $(\tau, \beta) - (0, \beta) = (\tau, 0)$. Consequently

$$A_2^* = 2A_2 - A_1 = (2\tau, -\beta), \quad A_2^* - A = (\tau, -2\beta).$$

2.2 The Pythagorean relation

Since O, A_1, A form a right angle at A_1 , we obtain directly

$$S = |OA|^2 = \tau^2 + \beta^2 = |AA_1|^2 + |OA_1|^2.$$

This is what the recurring block S actually is. Identity (iii) of §1.1, $|A_1A_2| = |OA|$, is also immediate in this frame: $A_2 - A_1 = (\tau, -\beta)$.

2.3 @ is a single angle

Setting $\theta = \text{atan2}(\beta, \tau)$ gives $A = |OA|(\cos \theta, \sin \theta)$, and

$$@ = \frac{|\beta|}{|\tau|} = |\tan \theta|.$$

Hence the branch condition $@ < 1$ is the same as $|\theta| < 45^\circ$, while $@ = 1$ corresponds to $\theta = \pm 45^\circ$ or $\pm 135^\circ$. Indeed the two $@ = 1$ examples used for verification give exactly $\theta = -45.00^\circ$ and 135.00° .

3. $A_{3a} \sim A_6$ — Eliminating Nine Points

The construction in the source text begins where the normal at A meets the two axes and passes through four line intersections before reaching A_{5a} and A_{5b} . Eliminating these nine points is the goal of this chapter. This part alone cannot be moved into the rotated frame, because the construction refers directly to the x -axis and the y -axis. That is also the sole reason why $Z1$ and $Z2$ differ at all.

3.1 Algebraic form of the construction

The normal at A has slope $-1/a$ and passes through A . Its intersections with the two axes are

$$A_{3a} = (t, 0), \quad A_{3b} = (0, y_b), \quad y_b = ap + b + \frac{p}{|a|}.$$

The denominator in A_{3b} is $|a|$, not a . This is the original construction rule as given; it is not something introduced or adjusted in the present derivation. For $a > 0$ it agrees with p/a and the difference is invisible, but for $a < 0$ the sign changes and the entire b -branch shifts position. Chapter 7 confirms this with an $a < 0$ example.

$$\begin{aligned} A_{3c} &= (t, at + b), & A_{3d} &= \left(\frac{y_b - b}{a}, y_b\right), & A'_{3a} &= (t, H), & A'_{3b} &= (-aH, y_b) \\ A_{4a} &= OA_{3c} \cap AA'_{3a}, & A_{4b} &= OA_{3d} \cap AA'_{3b} \end{aligned}$$

3.2 Reduction to scalars on line (1)

By the definitions in the source text, A_{5a} and A_{5b} lie on the same line (1) — the line through A_2 perpendicular to A_1A_2 . Each point is therefore specified by a single scalar along that line. Arranging the direction vector,

$$v = C(A_2 - A_1) = (t + ab, at - b), \quad w = \text{rot}90(v) = (b - at, t + ab).$$

Writing $A_{5a} = A_2 + s_a w$ and $A_{5b} = A_2 + s_b w$ and extracting s by symbolic computation gives

$$s_a = \frac{H}{Ct + ab}.$$

For $a > 0$ the denominator of s_b is the exact conjugate of that of s_a :

$$s_b = \frac{a^2 H}{Ct - ab} \quad (a > 0).$$

Multiplying the two denominators yields the single denominator $(Ct)^2 - (ab)^2$. All the information carried by the nine points was contained in the sum and difference of Ct and ab .

3.3 Passing to τ and β

Substituting $t = \tau\sqrt{C}$ and $b = \beta\sqrt{C}$ cancels every $\sqrt{C}$ and shortens the expression further. Writing $\lambda = C\sigma$ with $\sigma = s_a + s_b$,

$$\lambda = \frac{\beta}{C\tau + a\beta} + \frac{a^2\beta}{C\tau - a\beta} \quad (a > 0).$$

Written with $\varphi = \arctan a$ it becomes clear why the two terms are a pair.

$$Cs_a = \frac{\beta \cos^2 \varphi}{\tau + \beta \sin \varphi \cos \varphi}, \quad Cs_b = \frac{\beta \sin^2 \varphi}{\tau - \beta \sin \varphi \cos \varphi} \quad (a > 0)$$

The split is exactly into $\cos^2 \varphi$ and $\sin^2 \varphi$. The a -branch (which passes through the x -axis) and the b -branch (through the y -axis) carry as weights the squared cosine and squared sine of the tangent angle respectively, and their denominators differ only in the sign of $\beta \sin \varphi \cos \varphi$. This symmetry results from the source construction treating the two axes on an equal footing.

3.4 The symmetry breaks for $a < 0$

With $\varepsilon = \text{sgn}(a)$ one has $p/|a| = \varepsilon p/a$, so a factor ε attaches wherever p/a occurs in §3.1. Redoing the symbolic computation with this yields a unified form.

$$s_b = \frac{ab[aS + bp(1 - \varepsilon)]}{C\left\{tS(a^2 + \varepsilon) - ab[\varepsilon S + (\varepsilon - 1)pt]\right\}}$$

For $\varepsilon = 1$ the factor S cancels and one returns to §3.2. For $\varepsilon = -1$ it survives. Transferred to τ, β the result is

$$a < 0 : \quad Cs_b = \frac{a\beta(a\tau^2 + 2\beta\tau - a\beta^2)}{a^2\tau(\tau^2 - \beta^2) + a\beta(3\tau^2 + \beta^2) - \tau S}$$

Interpretation of the result. For $a > 0$ the conjugate pair $C\tau + a\beta$ and $C\tau - a\beta$ appears; for $a < 0$ that symmetry breaks and $S = |OA|^2$ enters the denominator instead. This is a consequence that follows from the $|a|$ rule in A_{3b} ; $|a|$ was not chosen in order to break the symmetry. The construction rule comes first, and the asymmetry emerges only when that rule is developed algebraically.

It may be added that for $a < 0$ both numerator and denominator are irreducible over the rationals. Attempts at factorisation by symbolic computation do not split them, so no simpler rational expression exists.

3.5 A_6 — a displacement from the reflection of A_1 in A_2

The source definition is $A_1 A_6 = A_1 A_{5a} + A_1 A_{5b}$, that is $A_6 = A_{5a} + A_{5b} - A_1$. Substituting,

$$A_6 = (2A_2 - A_1) + \sigma w = A_2^* + \sigma w, \quad A_2^* = 2A_2 - A_1.$$

Here A_2^* is the reflection of A_1 in A_2 . By identity (iii) one has $|A_1 A_2^*| = 2|OA|$, and moreover, independently of the value of σ , $(A_6 - A_2) \cdot (A_2 - A_1) = |OA|^2$ holds, because w is perpendicular to $A_2 - A_1$.

In the rotated frame the expression has only three components. By §2.1, $A_2^* - A = (\tau, -2\beta)$, and $w = \text{rot}90(C(A_2 - A_1)) = C(\beta, \tau)$, so $\sigma w = \lambda(\beta, \tau)$. Therefore

$$AA_6 = (\tau + \lambda\beta)T + (\lambda\tau - 2\beta)E$$

When $\lambda = 0$ this is $\tau T - 2\beta E$, which is precisely A_2^* . Thus λ is the single gauge measuring how far A_6 is displaced from that reference point. The ten points $A_{3a}, A_{3b}, A_{3c}, A_{3d}, A'_{3a}, A'_{3b}, A_{4a}, A_{4b}, A_{5a}, A_{5b}$ have all disappeared.

This is the formula corrected in the present edition; see the Erratum. The E -component previously carried the opposite sign.

4. $A_7 \sim A_{12}$ — Rotation Invariance

4.1 The key observation

Reading the source specification from A_7 onward, the objects that appear are O, A, A_1 , circle (1), circle (2), and the operations of perpendicularity, midpoint and rotation. Neither the x -axis nor the y -axis appears even once. Therefore the entire $A_7 \sim A_{12}$ construction is invariant under rotation about the origin.

But by §2.1, in the rotated frame $A_1 = (0, \beta)$ and $A = (\tau, \beta)$, which is exactly the configuration “ $y = c, A = (c_1, c)$ ” of Z2 in the source text. Hence

$$A_7 \sim A_{12} (Z1) = A_7 \sim A_{12} (Z2, y = c) \text{ computed with } c = \beta, c_1 = \tau, \text{ then rotated back.}$$

The stipulation in the source text that in Z2 “the computation from A_7 onward is carried out in exactly the same way as in Z1” is a reflection of this invariance. The present derivation shows that the converse direction — rotating Z1 into Z2 — holds as well. Chapter 7 confirms this numerically for ten parameter combinations.

4.2 A_7 — circle (1) and the line OA_1

The radius of circle (1) is $R = |AA_1| = |\tau|$, and the unit vector in the direction $O \rightarrow A_1$ is $\text{sgn}(\beta)E$. Since A_7 lies a further R beyond A_1 ,

$$\text{rotated frame: } A_7 = (0, \beta + \text{sgn}(\beta)|\tau|), \quad \text{original: } A_7 = \frac{b + \text{sgn}(b)|t|}{C} (-a, 1).$$

4.3 A_8 — reflection along a chord

A_8 is the second intersection of circle (1) with the line through A perpendicular to OA . Since A already lies on the circle, the second intersection follows at once from the reflection formula. The

direction is $\text{rot90}(A)$, whose squared norm is S .

$$\text{rotated frame: } A_8 = \left(\frac{\tau(\tau^2 - \beta^2)}{S}, \frac{\beta(3\tau^2 + \beta^2)}{S} \right), \quad \text{original: } A_8 = A + \frac{2tb}{CS}(-(ap + b), p).$$

Membership in circle (1) and $AA_8 \perp OA$ were confirmed symbolically (residual 0).

4.4 A_9 and A_{9a} — inside/outside test as a single sign

The source text takes the midpoint M of segment A_7A_8 and the two points at perpendicular distance $|A_7A_8|/2$ from it, calling the one inside circle (1) A_9 and the one outside A_{9a} . Both candidates are written immediately with rot90 .

$$M = \frac{A_7 + A_8}{2}, \quad d = A_7 - A_8 \Rightarrow \text{candidates} = M \pm \frac{1}{2} \text{rot90}(d)$$

Which one lies inside can be decided without computing any distance. Since A_7 and A_8 both lie on the circle, $|M - A_1|^2 = R^2 - h^2$ with $h = |d|/2$, and the squared distance to a candidate is

$$(R^2 - h^2) + h^2 \pm 2hn \cdot (M - A_1) = R^2 \pm 2hn \cdot (M - A_1).$$

The h^2 cancels, so inside versus outside is decided by one sign.

$$\varepsilon_9 = -\text{sgn}(\text{cross}(d, M - A_1)), \quad A_9 = M + \frac{\varepsilon_9}{2} \text{rot90}(d), \quad A_{9a} = M - \frac{\varepsilon_9}{2} \text{rot90}(d)$$

4.5 A_{10} — second intersection with circle (2)

The two branches of the source text unify into an exchange of centre and base point.

branch	centre Q of circle (2)	base point P	radius
$\mathfrak{Q} < 1$	A_8	A_9	$ A_8A_9 $
$\mathfrak{Q} > 1$	A_9	A_7	$ A_9A_7 $

Table 1. The two branches of circle (2). In either case the base point P lies on circle (2).

A_{10} is the intersection of circle (2), other than P , with the line through P perpendicular to AP , so the same reflection formula as in §4.3 applies.

$$n = \text{rot90}(P - A), \quad A_{10} = P + 2 \frac{(Q - P) \cdot n}{n \cdot n} n$$

4.6 Eliminating A'_{10}

The source text defines A_{11} by way of A'_{10} , which is the intersection of the line AP with “the line through A_{10} having the reciprocal slope of AA_{10} ”; A_{11} is then the point taken from A in the direction $A'_{10} \rightarrow A_{10}$ at distance $|AA_{10}|$.

Observe the condition $A_{10}A'_{10} \perp AA_{10}$. The direction $A'_{10} \rightarrow A_{10}$ is perpendicular to AA_{10} , and the length is prescribed to be $|AA_{10}|$; hence A_{11} is nothing but AA_{10} turned through 90 degrees. The only remaining question is which way it turns.

With $u = A_{10} - A$ and $r = P - A$ we have $A'_{10} = A + \lambda_0 r$, and the perpendicularity condition $(u - \lambda_0 r) \cdot u = 0$ gives $\lambda_0 = (u \cdot u)/(r \cdot u)$. Therefore

$$A_{10} - A'_{10} = u - \frac{u \cdot u}{r \cdot u} r.$$

Taking the inner product of this vector with $\text{rot90}(u)$ kills the first term, since $u \cdot \text{rot90}(u) = 0$, and only a sign survives.

$$A_{11} = A - \text{sgn}(\text{cross}(u, r) \cdot (u \cdot r)) \text{rot90}(u)$$

A_{11} is obtained exactly without ever computing A'_{10} . For verification purposes A'_{10} can be recovered by back-substituting $\lambda_0 = (u \cdot u)/(r \cdot u)$ — this is done against the plotted values in Chapter 7.

4.7 A_{12}

$$AA_{12} = AA_6 + AA_{11} \implies A_{12} = A_6 + A_{11} - A$$

5. Special Cases

5.1 $\mathcal{Q} = 1$ — collapse of circle (2)

If $|b| = |t|$ then $A_7 = A_8$, hence $d = 0$ and $A_9 = A_{9a} = A_7$; the radius of circle (2) becomes 0 and $A_{10} = A_7$. The sign expression of §4.6 then breaks down as $0/0$, because $u \parallel r$ gives $\text{cross}(u, r) = 0$. The source text prescribes this case separately: rotate AA_7 through 90 degrees in the same sense in which AO rotates toward AA_7 by at most 90 degrees.

$$\mathcal{Q} = 1 : \quad A_{11} = A + \text{sgn}(\text{cross}(AO, AA_7)) \text{rot90}(AA_7)$$

$|AA_{11}| = |AA_7|$ and $AA_{11} \perp AA_7$ were verified numerically. By §2.3 this case corresponds to $\theta = \pm 45^\circ$ or $\pm 135^\circ$.

5.2 $t = 0$ — A coincides with A_1

If $t = Cp + ab = 0$ then by (i) $A = A_1$ and $|AA_1| = 0$, so circle (1) collapses to a point. Since $\mathcal{Q}$ is undefined and A_{11} cannot be generated,

$$t = 0 : \quad AA_{12} = AA_6, \quad \text{that is} \quad A_{12} = A_6.$$

The denominator $(Ct)^2 - (ab)^2$ of σ equals $-(ab)^2$ at $t = 0$, which is non-zero, so A_6 itself is computed normally. The source rule “ $y = ax$, $A(x, ax) \rightarrow A_{12} = A$ ” is the sub-case $b = 0$, where $\beta = 0$ forces $\lambda = 0$ and $A_6 = A$, giving the self-returning structure.

5.3 $Z2$ — reference line $x = k$ or $y = c$

These are the cases where the slope is undefined or zero. By §4.1 everything from A_7 onward is identical to $Z1$, and only A_1 , A_2 and A_6 need replacing.

Case $y = c$, $A = (c_1, c)$ — this corresponds to $Z1$ with $a = 0$, $b = c$, $p = c_1$. Substituting gives $C = 1$, $H = c$, $t = c_1$, $S = c_1^2 + c^2$, and

$$A_1 = (0, c), \quad A_2 = (c_1, 0), \quad \mathcal{Q} = \frac{|c|}{|c_1|},$$

$$A_7 = (0, c + \text{sgn}(c)|c_1|), \quad A_8 = \left(\frac{c_1(c_1^2 - c^2)}{S}, \frac{c(3c_1^2 + c^2)}{S} \right).$$

A_6 , however, is not the $a \rightarrow 0$ limit. Since A_{3b} of $Z1$ contains $p/|a|$, it diverges at $a = 0$; the source text therefore gives A_6 in $Z2$ by a separate construction rule.

Construction rule (source). “Intersection of the line through A_2 perpendicular to A_1A_2 with the vertical/horizontal line of the reference line. $x = k \rightarrow$ vertical line, $y = c \rightarrow$ horizontal line.”

Meaning of the rule. Read as a sentence alone, this wording leaves some ambiguity as to which line is meant. According to the actual construction and the values in plot ①, the vertical/horizontal line here is the reference line itself. That is, A_6 is the intersection of line (1) with the reference line. What follows develops that reading, and Chapter 7 confirms agreement with plot ① to five decimal places.

Since $A_1 = (0, c)$ and $A_2 = (c_1, 0)$, line (1) has direction $\text{rot90}(c_1, -c) = (c, c_1)$; substituting $y = c$ gives parameter c/c_1 .

$$y = c : \quad A_6 = \left(c_1 + \frac{c^2}{c_1}, c \right)$$

Case $x = k$, $A = (k, k_1)$ — no separate computation is needed. It is the reflection of the case $y = c$ in the line $y = x$, and the entire Sangbun construction is invariant under this reflection: none of its selection rules (the point inside the circle, the other intersection, the direction $A'_{10} \rightarrow A_{10}$) refers to handedness.

$$Z2(x = k, A = (k, k_1)) = \text{swap} \left[Z2(y = c = k, A = (c_1 = k_1, c = k)) \right]$$

$$A_1 = (k, 0), \quad A_7 = (k + \text{sgn}(k)|k_1|, 0), \quad A_6 = \left(k, k_1 + \frac{k^2}{k_1} \right), \quad \mathfrak{Q} = \frac{|k|}{|k_1|}$$

6. Computational Procedure (all cases unified)

step	expression
0. setup	$C = a^2 + 1, \quad H = b/C, \quad t = Cp + ab, \quad S = p^2 + (ap + b)^2$ $\tau = t/\sqrt{C}, \quad \beta = b/\sqrt{C}, \quad T = (1, a)/\sqrt{C}, \quad E = \text{rot90}(T)$
1. base points	$A = (p, ap + b), \quad A_1 = H(-a, 1), \quad \mathfrak{Q} = b / t = \beta / \tau $
2. λ	$\lambda = \frac{\beta}{C\tau + a\beta} + (\text{sign-dependent term of §3.3 or §3.4})$
3. A_6	$AA_6 = (\tau + \lambda\beta)T + (\lambda\tau - 2\beta)E$ [corrected]
4. A_7, A_8	$A_7 = [(b + \text{sgn}(b) t)/C](-a, 1)$ $A_8 = A + [2tb/(CS)](-(ap + b), p)$
5. A_9	$M = (A_7 + A_8)/2, \quad d = A_7 - A_8$ $\varepsilon_9 = -\text{sgn}(\text{cross}(d, M - A_1)), \quad A_9 = M + (\varepsilon_9/2) \text{rot90}(d)$
6. A_{10}	choose P, Q from Table 1; $n = \text{rot90}(P - A)$ $A_{10} = P + 2[(Q - P) \cdot n / (n \cdot n)]n$
7. A_{11}	$u = A_{10} - A, \quad r = P - A$ $A_{11} = A - \text{sgn}(\text{cross}(u, r) \cdot (u \cdot r)) \text{rot90}(u)$
8. A_{12}	$A_{12} = A_6 + A_{11} - A$
exceptions	$t = 0 \rightarrow A_{12} = A_6 \mid \mathfrak{Q} = 1 \rightarrow \text{§5.1} \mid Z2 \rightarrow \text{§5.3}$

Table 2. No line-intersection solving occurs anywhere; only arithmetic and sign functions.

7. Verification

All numbers in this chapter were recomputed from scratch with the corrected formula of §3.5.

7.1 Comparison with the plots

plot	conditions	item	printed on plot	this formulation
①	$Z2, x = 5$	A_{12}	$(0.58824, -9.74510)$	$(0.58824, -9.74510)$
	$A = (5, -3)$	$ AA_{12} $	8.05978	8.05978
	$\mathfrak{Q} = 1.66667$	angle	-123.19°	-123.19°
		$ A_1A_6 $	11.33333	11.33333
		A'_{10}	$(8.66436, 0.66436)$	$(8.66436, 0.66436)$
		$ A'_{10}A_{11} $	8.33871	8.33871
②	$Z1, a = 3/5$	A_1	$(-2.64706, 4.41176)$	$(-2.64706, 4.41176)$
	$b = 6, p = 6$	A_{12}	$(28.57893, 20.61668)$	$(28.57893, 20.61668)$
	$\mathfrak{Q} = 0.510204$	A'_{10}	$(-10.25565, 11.41313)$	$(-10.25565, 11.41313)$
③	$Z1, a = 1/2$	A_{12}	$(4.80000, -0.60000)$	$(4.80000, -0.60000)$
	$b = 3, t = 0$	$ AA_{12} $	6.70820	6.70820
	$A = A_1$	angle	-26.57°	-26.57°
④	$Z1, a = -3/5$	A_{12}	$(19.42977, -32.71909)$	$(19.42977, -32.71909)$
	$b = -6$	$ AA_{12} $	31.15291	31.15291
	$\mathfrak{Q} = 1$	angle	-55.46°	-55.46°
		$ A_1A_6 $	29.39526	29.39526
		$ A'_{10}A_{11} $	10.28992	10.28992

Table 3. Every item agrees to five decimal places. Plot ④ has $a < 0$ and verifies the $|a|$ rule in A_{3b} . Plots ② and ④, which involve A_6 , are reproduced only with the corrected sign of §3.5.

7.2 Closed form against the full construction

A_6 was computed (a) by carrying out the entire $A_3 \sim A_5$ construction as specified in the source text and (b) by the closed form of §3.5, and the two were compared.

a	b	p	max. error in A_6
0.6	6	6	3.6×10^{-15}
-0.6	-6	1.7647058824	1.4×10^{-14}
-1.5	4	7	1.6×10^{-12}
2	-1	0.3	1.5×10^{-14}
-0.7	-2	2.5	1.8×10^{-15}
3	1	-0.4	2.2×10^{-16}
-2	5	-5	0
0.2	0.5	-3	8.9×10^{-16}
-4	-3	1.1	1.8×10^{-15}
1	1	1	8.9×10^{-16}

Table 4. Errors at the level of floating-point rounding. Both signs of a are included.

7.3 Confirmation of rotation invariance

The claim of §4.1 — that computing $A_7 \sim A_{11}$ by the $Z2$ ($y = c$) formulas in the rotated frame and rotating back reproduces the $Z1$ values — was checked for the same ten combinations.

a	$\beta (= b/\sqrt{C})$	$\tau (= t/\sqrt{C})$	θ	$\mathcal{Q}$	max. error $A_7 \sim A_{11}$
0.6	5.145	10.084	27.03°	0.5102	7.1×10^{-15}
-0.6	-5.145	5.145	-45.00°	1.0000	5.3×10^{-15}
-1.5	2.219	9.291	13.43°	0.2388	3.6×10^{-15}
2	-0.447	-0.224	-116.57°	2.0000	1.4×10^{-16}
-0.7	-1.638	4.199	-21.32°	0.3902	8.9×10^{-16}
3	0.316	-0.316	135.00°	1.0000	3.3×10^{-16}
-2	2.236	-15.652	171.87°	0.1429	3.6×10^{-15}
0.2	0.490	-2.961	170.60°	0.1656	8.9×10^{-16}
-4	-0.728	7.446	-5.58°	0.0977	1.8×10^{-15}
1	0.707	2.121	18.43°	0.3333	8.9×10^{-16}

Table 5. Here $\theta = \text{atan2}(\beta, \tau)$; the relation $\mathcal{Q} = |\tan \theta|$ can also be read off.

7.4 Auxiliary conditions

Maximum residual over the ten combinations of Table 4:

- A_7 and A_8 lie on circle (1) — residual 1.8×10^{-15}
- A_9 inside circle (1), A_{9a} outside, and $|A_7 A_9| = |A_8 A_9|$
- A_{10} lies on circle (2) — residual 1.8×10^{-15}
- $AA_{11} \perp AA_{10}$ and $|AA_{11}| = |AA_{10}|$ — residual 1.8×10^{-14}
- $S = \tau^2 + \beta^2$ (Pythagoras) — residual 2.8×10^{-14}
- $(A_6 - A_2) \cdot (A_2 - A_1) = |OA|^2$ — residual 5.7×10^{-14}
- $|A_1 A_2^*| = 2|OA|$ — residual 1.8×10^{-15}
- the closed form for AA_6 was confirmed symbolically to be identical with the construction — residual exactly 0

7.5 The limiting-direction law revisited

The previously discovered law — that as $p \rightarrow \pm\infty$ the direction of AA_{12} deviates from the tangent angle by exactly 45 degrees — was re-confirmed with the new closed form. It holds regardless of the sign and magnitude of a .

a	$\arctan a$	deviation, $p \rightarrow +\infty$	deviation, $p \rightarrow -\infty$
0.6	30.964°	45.000°	135.000°
-0.6	-30.964°	45.000°	135.000°
3	71.565°	45.000°	135.000°
-1.5	-56.310°	45.000°	135.000°
-10	-84.289°	45.000°	135.000°

Table 6. Measured with $b = 3$ fixed and $|p| = 10^7$.

7.6 Plot ② computed in full: $a = 3/5$, $b = 6$, $x = 6$

Plot ② of Table 3 prints only the three points A_1 , A_{12} and A'_{10} . Here the same conditions are carried through every step of the procedure of Table 2. It is a $Z1$ case with $a > 0$ in the $\textcircled{< 1}$ branch; A_6 was cross-checked against the full $A_3 \sim A_5$ construction (agreement 3.6×10^{-15}).

quantity	value	quantity	value
C	1.36	A	(6.00000, 9.60000)
H	4.41176	A_1	(−2.64706, 4.41176)
t	11.76000	A_2	(8.64706, 5.18824)
S	128.16000	A_6	(19.56808, 11.39160)
τ	10.08412	A_7	(−7.83529, 13.05882)
β	5.14496	A_8	(−1.77264, 14.45790)
θ	27.03°	A_9	(−4.10443, 10.72703)
λ	0.48051	A_{9a}	(−5.50350, 16.78969)
$\textcircled{<}$	0.51020 ($\textcircled{<} < 1$)	A_{10}	(−3.22508, 18.61085)
$ A_1 A_6 $	23.28584	A'_{10}	(−10.25565, 11.41313)
$ A'_{10} A_{11} $	26.33122	A_{11}	(15.01085, 18.82508)
$ A A_{12} $	25.12320	A_{12}	(28.57893, 20.61668)
angle of AA_{12}	26.01°		

Table 8. All quantities for the conditions of plot ②, five decimal places. Branch: circle (2) has centre A_8 , radius $|A_8 A_9|$, base point $P = A_9$. The entries A_1 , A_{12} and A'_{10} are those that appear on the plot itself.

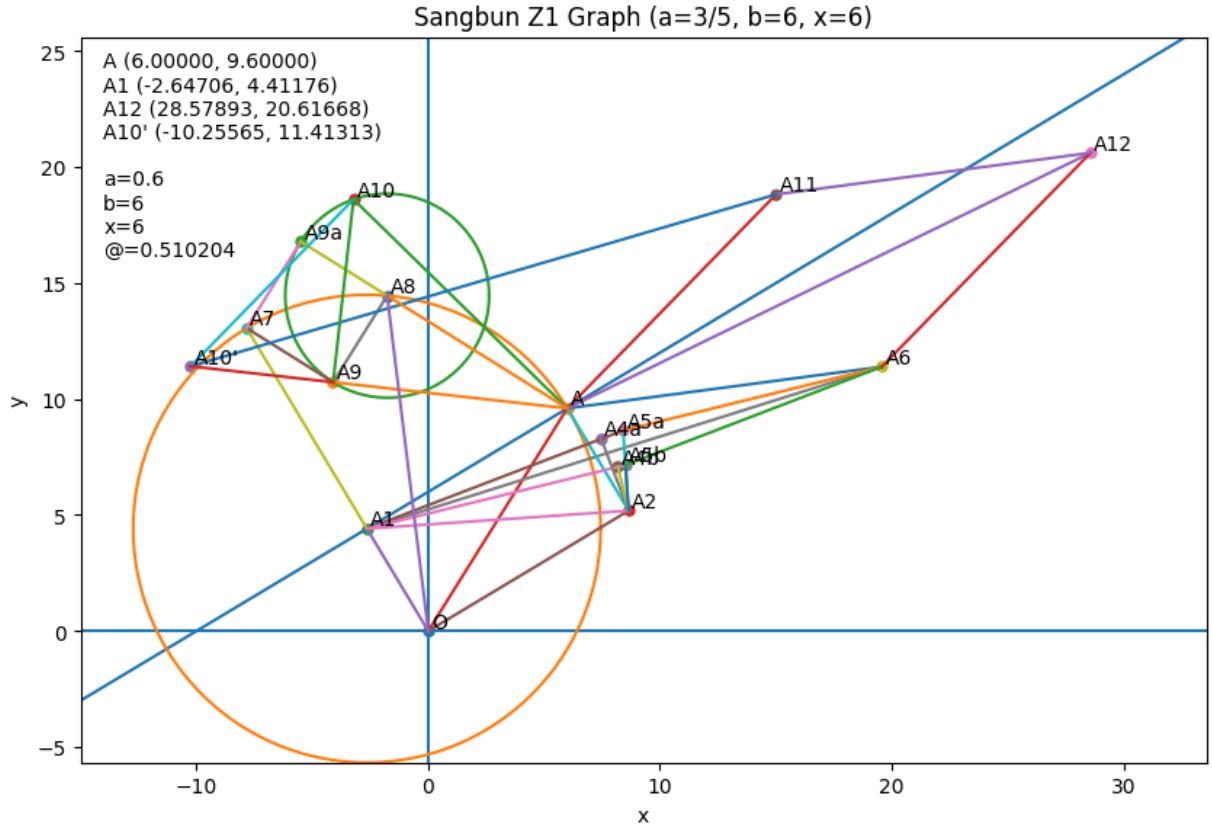


Figure 1. The plot produced by the existing construction program for the conditions of Table 8. Axis ratio 1:1; as prescribed, coordinates are printed only for A , A_1 , A_{12} and A'_{10} . All thirteen points of Table 8 agree with the positions drawn here. The large circle is circle (1) (centre A_1 , radius $|A_1A|$) and the small one is circle (2) (centre A_8 , radius $|A_8A_9|$). The points A_{4a} , A_{4b} , A_{5a} and A_{5b} , which the procedure of Table 2 eliminates, are also shown here as a check on the construction.

8. Summary

what was eliminated	what replaces it
A_{3a} , A'_{3a} , A_{3b} , A'_{3b} , A_{3c} , A_{3d}	(eliminated entirely)
A_{4a} , A_{4b} , A_{5a} , A_{5b}	a single scalar λ
A'_{10}	the sign $\text{sgn}(\text{cross}(u, r) \cdot (u \cdot r))$
the square root in $@$	comparison of $ b $ and $ t $, or the angle θ
distance test for A_9 inside/outside	the sign $-\text{sgn}(\text{cross}(d, M - A_1))$
four simultaneous line-intersection systems	two reflection formulas
the Z1 / Z2 dichotomy	rotation invariance (identical from A_7 on) + A_6 only

Table 7. Summary of the compression.

A twelve-step construction has been reduced to arithmetic together with four sign functions. The only branches remaining are the three cases of $@$, the sign of a , and $t = 0$, each decided by a single comparison.

Three structural facts have emerged. First, the Sangbun construction splits into two parts: up to A_6 it refers to the coordinate axes and therefore depends on rotation, whereas from A_7 on it

is rotation-invariant. The source stipulation that in $Z2$ “everything from A_7 on is exactly as in $Z1$ ” reflects precisely this fact. Second, that rotation-invariant part is completely determined by the two lengths $\tau (= |AA_1|)$ and $\beta (= |OA_1|)$, and the branch value $\mathfrak{Q}$ is the tangent of the angle they subtend. Third, the $p/|a|$ in A_{3b} is the original construction rule, and the $a < 0$ asymmetry that follows from it appears explicitly in the formulas for the first time in this derivation: for $a > 0$ the conjugate symmetry $C\tau + a\beta$ versus $C\tau - a\beta$ holds and the weights split into $\cos^2 \varphi$ and $\sin^2 \varphi$, while for $a < 0$ that symmetry breaks and S enters the denominator. It should be stated plainly that the rule produced the result, not the reverse.

Appendix · verification environment. Symbolic computation was carried out with SymPy and numerical verification with NumPy in double precision. A'_{10} was recovered by back-substituting $\lambda_0 = (u \cdot u)/(r \cdot u)$ and compared with the plotted values — it is not used in the computational path but remains valid for verification. Chapter 0 reproduces the author’s source text; the derivations, interpretations and naming from Chapter 1 onward are added in the present document.

Note on this edition. The sole mathematical change relative to the first edition is the sign of the E -component of AA_6 in §3.5 and Table 2, together with the explicit statement $A_2 = (\tau, 0)$ in §2.1 from which it follows. All verification tables were regenerated after the correction. The source text of Chapter 0 is unchanged.